

CS250-F26-Group03: Casino Blackjack Cheater Detection System

I. Project Description and Objective

A casino wants a system that helps its surveillance team spot possible card counting and other behavior that may need investigation. Card counters keep track of the high- and low-value cards, or favorable cards, already revealed. They use that information to judge whether the remaining cards favor the player and may adjust their bets or decisions. Since a winning streak or one smart move is not enough evidence to determine whether someone is counting cards, our system will collect and analyze data from equipped blackjack tables and synchronized cameras. It will rebuild each round and look for patterns over time, while staff review any concerning activity. Card counting alone is not proof of fraud or illegal cheating, so the system will show the evidence and how confident it is. Casino staff will decide whether any action is allowed under casino policy and applicable law.

II. Physical Equipment and Data Collection

- **Seat sensors**

Each active player seat has a built-in card-reading sensor. The sensor identifies the rank and suit of each dealt card. It records which seat and hand received the card, along with the time and how confident the reading is. If a reading is unclear, the system checks the camera footage or dealer feed.
- **Table and Dealer Inputs**

A dealer-side reader records the dealer's cards and which cards are publicly visible. It also tracks shuffles and the number of decks in play. A camera or table sensor records wagers. Synchronized footage captures hits and stands, plus doubles, splits and surrenders when allowed.
- **Surveillance Cameras**

Overhead and table-facing cameras provide synchronized footage of the game. They record dealt cards and wagers, along with player actions and seat changes. They also capture interruptions. Staff use footage to check uncertain sensor readings and review alerts; video alone cannot prove misconduct.
- **Shoe Sequence**

An approved card-reading shoe records the verified order of cards after each shuffle. The system keeps this order in a restricted area for card-sequence checks and separate investigations. The system's card-counting analysis cannot assume that a player knows upcoming cards or the dealer's hidden card.

III. Detection Workflow

1. Reconstruct Each Round

The system links each card and action to the correct table and seat. It assigns wagers and hands to an anonymous session. It checks duplicate events and resolves conflicting sensor and camera readings.

2. Estimate Public Card Information

The system only uses the cards visible when a player makes a decision to determine a count. The purpose of the count is to track a net value of favorable and unfavorable cards on the table. It continues to keep track of this configurable running count and estimates how many decks remain, then calculates the true count. This estimates what the player could know from exposed cards, and how favorable the player's decision is. The count resets after a shuffle or shoe replacement.

3. Evaluate Decisions and Wagering

The strategy module uses table rules to find the basic-strategy move for each hand against the dealer's visible card. It can also model decisions that change with the count. The analyzer compares these moves and betting patterns with the player's actions across several valid rounds.

4. Generate An Evidence-Based Alert

The risk model looks for bets and decisions that repeatedly follow the count. It considers the number of rounds and the reliability of the data. It also checks for other explanations of fraudulent play since one matching move, large bet or winning hand is not enough to accuse a player.

5. Require Human Review

Operators review video and sensor readings, then check the decision timeline. A supervisor can confirm the need for follow-up, dismiss the case or request more observation. Every review is logged.

IV. System Users and Access

1. An operator watches authorized live feeds and reviews alerts. The operator checks session timelines and related clips, reviews confidence levels and records an initial assessment.

2. A pit supervisor reviews escalated cases and checks table conditions. The supervisor chooses any response allowed under casino policy. The system never removes or penalizes players, or publicly identifies them.

3. A system administrator configures table equipment, blackjack rules and alert thresholds. The administrator also manages staff permissions and oversees equipment health and maintenance.

4. An auditor has read-only access to approved event history and incident records. The auditor can also see review decisions and changes to system settings.

V. Required System Capabilities and UI Functionality

● Table Configuration

Staff set deck count, betting limits and the seat layout. They also configure the dealer's soft-17 rule and the options for doubling, splitting or surrendering. The system records all shuffles and activity from continuous shuffling machines.

● Real-time Monitoring

The dashboard shows table status and active seats. It further details reconstructed hands, exposed cards, bets and player actions. Staff check reading confidence and device health. The system handles split hands, seat changes and interrupted rounds.

● Analysis

The system compares each session's bets and decisions with basic-strategy and card-counting models. Staff set minimum sample sizes and alert thresholds. Missing or unreliable readings lower confidence or pause scoring; the system never guesses hidden cards.

- Alerts and Review

Each alert shows the relevant table, seat and time period. It includes the actions that raised concern, the count and betting trend, a confidence level and linked video. Operators can mark an alert as pending, under review, confirmed for follow-up or dismissed. Each decision includes notes and the reviewer's identity.

- Recordkeeping and Reporting

Staff can search incidents by table, session, time or review outcome. Reports summarize sensor failures and alert volumes. They also show which alerts were confirmed as false positives. The system records evidence access and any changes to stored records.

VI. Reliability, Privacy, and Security

- Operational Reliability

Each event requires an allocated unique ID and a synchronized timestamp. When local buffering is available, collection should continue through brief network outages. The dashboard must warn staff about unavailable devices or incomplete rounds.

- Security and Privacy

Staff sign in to authenticated accounts with access based on their roles. The system encrypts data in transit and in storage. It logs access and follows a casino-approved retention schedule that meets applicable requirements. Sessions stay anonymous unless identifying a player is separately authorized and necessary.

- False-positive Controls

Good basic-strategy play does not prove misconduct. Neither does a winning streak or a single betting change. Before scoring, the system checks the rules and how much of the shoe has been dealt to further determine the validity of the alert. It also accounts for shuffles and seat changes and corrects sensor errors. Staff must review every alert.

VII. Initial Implementation and Boundaries

The pilot will cover one blackjack table with up to seven player seats. It will include dealer inputs, table sensors and two synchronized camera views. A local service will process events, while a dashboard supports review and a database stores audit records. After testing with recorded sessions, the system should scale to more tables without changing the core analysis. Proposed goals are processing valid events within two seconds and showing qualified alerts within ten seconds under normal connectivity. The client will set accuracy targets and acceptable false-positive rates using representative test data.

Boundaries: The system flags patterns for review; it cannot prove cheating or identify a player by itself. Its count analysis cannot use hidden information available only to the casino. It cannot take disciplinary action without staff approval. If a player's choices suggest knowledge of upcoming cards, staff can open a separate investigation. That possibility must not be confused with ordinary card-counting evidence.